

Multi-Level Richardson Extrapolation for Faster Diffusion ODE Sampling

Group 10 · Section A · CSE 402: Numerical Analysis, Simulation and Modeling Sessional

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Abstract

Diffusion probabilistic models generate samples by numerically solving a probability flow ODE, so the quality of a sample under a fixed budget of network function evaluations (NFEs) is governed by the truncation error of the solver. RX-DPM reduces that error by Richardson extrapolation: two numerical solutions of the same time interval, obtained on different discretizations of it, are linearly combined with coefficients derived from the local truncation error of the baseline solver on a non-uniform grid. The leading error term is cancelled without additional NFEs, because the coarse solution reuses a denoised prediction that the fine solution has already computed. We reproduce RX-DPM on CIFAR-10 with the EDM backbone and extend it from two levels to L nested levels, in the manner of a Romberg tableau, by solving a generalized Vandermonde moment system for the extrapolation weights. On an analytically solvable diffusion ODE, for which the exact solution is available, we find that the extension does not attain the predicted order of accuracy. The reuse of stored predictions, which is precisely what makes the two-level scheme free, is exact only at two levels, and for every $L > 2$ it limits the attainable local order to approximately 3.2. Recomputing each level honestly restores the predicted order but costs more evaluations than it returns. The weight system also becomes ill conditioned at a rate of roughly one order of magnitude per level. On CIFAR-10, RX-Euler lowers FID from 17.48 to 6.15 at $\text{NFE} = 10$, whereas three levels raise it to 66.3. Code is available at <https://github.com/azraihan/multi-level-richardson-faster-diffusion-ode>.

1 Introduction & Base Paper Review

Diffusion probabilistic models (DPMs) [5, 14] have become a standard framework for high quality generative modeling, but their sampling procedure is sequential and expensive. Generation is formulated as the solution of an ODE or SDE [14], and each step of the numerical solver requires one evaluation of a large neural network. The number of function evaluations (NFEs) is therefore the natural measure of sampling cost, and the truncation error of the solver is strongly correlated with the quality of the generated sample. Reducing that error without increasing the NFE is the central problem addressed in this work.

1.1 Base paper

Our base paper is *Enhanced Diffusion Sampling via Extrapolation with Multiple ODE Solutions* by Choi, Kang and Han, published at ICLR 2025 [2]. The method proposed there is referred to as RX-DPM.

The key ingredient of RX-DPM is Richardson extrapolation [11], a classical technique that combines numerical estimates obtained on grids of different resolution so that the leading term of the error expansion cancels. Standard Richardson extrapolation assumes a uniform discretization of the grid. Time step schedules used by DPMs are strongly non-uniform, since a finer spacing near the clean sample is known to be more beneficial for a fixed number of steps [6, 13], so the classical fixed coefficients are not appropriate. The authors therefore derive the truncation error of the Euler method on a non-uniform grid and obtain extrapolation coefficients that depend on the actual step widths of the schedule. They refer to these as grid-aware coefficients.

The algorithm is applied repeatedly over blocks of k denoising steps. Within each block, two numerical solutions of the same interval are available: a single-step solution and a k -step solution. Because the single-step solution requires only the denoised prediction at the start of the block, which the k -step solution has already computed and stored, the extrapolation requires no additional NFEs and incurs only the negligible cost of a linear combination of two tensors. The authors report consistent improvements in FID across EDM [6], Stable Diffusion V2, DPM-Solver [9] and PNDM [8] backbones, and provide an explicit error estimate showing that the order of the leading error term increases.

1.2 Objective of this work

Our work has two parts. First, we reproduce the core result of RX-DPM on CIFAR-10 with the EDM backbone, including the ablation against conventional Richardson extrapolation with fixed coefficients. Second, we extend the method along the direction that the classical theory suggests but the paper does not pursue. Romberg integration [12] does not stop at a single extrapolation; it stacks them into a tableau, where each successive column cancels one further term of the error expansion. We construct the analogous tableau on non-uniform grids, which requires solving a generalized Vandermonde system for the extrapolation weights rather than applying a closed-form ratio, and we examine what limits it.

The investigation is framed around three quantities that the course treats directly: the order of accuracy actually attained, the conditioning of the linear system that produces the weights, and the balance between truncation error and round-off error. Our contribution is a negative result with a precise and measurable cause. The extension does not attain the predicted order, because the mechanism that makes RX-DPM free of additional NFEs is exact only at two levels and does not survive generalization.

2 Methodology

2.1 Sampling as solving an ODE

Let $\mathbf{x} \in \mathbb{R}^d$ denote the state and let t denote the noise level. Following the EDM formulation of Karras et al. [6], in which the scaling is $s(t) = 1$ and the noise schedule is $\sigma(t) = t$, the probability flow ODE reduces to

$$d\mathbf{x} = \frac{\mathbf{x} - D_\theta(\mathbf{x}; t)}{t} dt, \quad (1)$$

where $D_\theta(\mathbf{x}; t)$ is the learned denoiser, that is, the network prediction of the clean sample given the noisy state $\mathbf{x}$ at noise level t . Sampling amounts to integrating Equation (1) from $t = T$, where $\mathbf{x}$ is approximately pure noise, down to $t = 0$. Given a time step schedule $0 = t_0 < t_1 < \dots < t_N = T$, the solution is obtained by applying a numerical solver over each subinterval, and one evaluation of D_θ is one NFE.

The schedule itself is the source of the non-uniformity. EDM places the noise levels at

$$t_{N-i} = \left(t_{\max}^{1/\rho} + \frac{i}{N-1} (t_{\min}^{1/\rho} - t_{\max}^{1/\rho}) \right)^\rho, \quad i = 0, \dots, N-1, \quad (2)$$

with $t_0 = 0$, so that $t_N = t_{\max}$ is the largest noise level and $t_1 = t_{\min}$ the smallest non-zero one. Here $t_{\min}$ and $t_{\max}$ bound the noise range and the exponent ρ controls how sharply the steps concentrate towards the clean end. Larger ρ shortens the steps near $t_{\min}$ and stretches those near $t_{\max}$; the value $\rho = 7$ is adopted by EDM on empirical grounds [6].

2.2 Truncation error on a non-uniform grid

Consider one block of k steps covering the interval $[t_{i-k}, t_i]$, and let

$$h = t_i - t_{i-k}, \quad \lambda_j = \frac{t_{i-j+1} - t_{i-j}}{h}, \quad j = 1, \dots, k, \quad (3)$$

so that λ_j are the normalized widths of the sub-steps and $\sum_{j=1}^k \lambda_j = 1$. Write $\hat{\mathbf{x}}_{t_{i-k}}^{(n)}$ for the numerical solution at t_{i-k} obtained with n iterations of the baseline solver from $\mathbf{x}_{t_i}$, and $\mathbf{x}_{t_{i-k}}^*$ for the exact solution. For a solver of local order p , the base paper [2] shows that the single-step and k -step solutions satisfy

$$\hat{\mathbf{x}}_{t_{i-k}}^{(1)} = \mathbf{x}_{t_{i-k}}^* + \mathbf{c} h^p + O(h^q), \quad (4)$$

$$\hat{\mathbf{x}}_{t_{i-k}}^{(k)} = \mathbf{x}_{t_{i-k}}^* + \mathbf{c} \sum_{j=1}^k \lambda_j^p h^p + O(h^q), \quad (5)$$

for $0 < p < q$ and a constant $\mathbf{c} \neq \mathbf{0}$. The Euler method, which is also the interpretation of DDIM [13] as an ODE solver, corresponds to $p = 2$. It is convenient to introduce the moments of the sub-step widths,

$$S_r(\Lambda) = \sum_{\lambda \in \Lambda} \lambda^r, \quad (6)$$

where Λ is the set of normalized widths of a given discretization of the block. The single-step discretization has $\Lambda = \{1\}$ and hence $S_r = 1$, while any finer discretization has $S_r < 1$ for $r > 1$. The leading error of a discretization is therefore controlled entirely by S_p .

Eliminating $\mathbf{c}$ between Equations (4) and (5) gives the extrapolated solution of RX-DPM,

$$\tilde{\mathbf{x}}_{t_{i-k}}^{(k)} = \frac{\hat{\mathbf{x}}_{t_{i-k}}^{(k)} - S_p \hat{\mathbf{x}}_{t_{i-k}}^{(1)}}{1 - S_p}, \quad (7)$$

which carries a truncation error of $O(h^q)$, asymptotically smaller than $O(h^p)$. The coefficients depend on the schedule through S_p alone, which is what distinguishes them from the fixed coefficients of conventional Richardson extrapolation.

Two orders are referred to in what follows and should be kept apart. The *local* order is the exponent of h in the error of a single block, as above. The *global* order is the exponent of N in the error of the complete trajectory, and is one lower, because a local error of $O(h^r)$ over a block of width $h = k/N$ accumulates over N/k blocks to $O(N^{-(r-1)})$. The base paper obtains $O(N^{-1})$ for the Euler method and $O(N^{-2})$ for RX-Euler in exactly this way. With $p = 2$, an L -level scheme should therefore reach local order $L + 1$ and global order L .

2.3 Extension to multiple levels

Nothing in the derivation restricts the construction to two discretizations. Let $\Lambda_1, \dots, \Lambda_L$ denote L discretizations of the same block, ordered from coarsest to finest, where Λ_n divides the block into k_n sub-steps, $k_1 = 1$ and $k_L = k$. Extending the linear error accumulation of Equation (5) to the subsequent terms of the expansion, we posit

$$\hat{\mathbf{x}}_{t_{i-k}}^{(k_n)} = \mathbf{x}_{t_{i-k}}^* + \sum_{m=1}^{L-1} \mathbf{b}_m S_{p+m-1}(\Lambda_n), \quad n = 1, \dots, L, \quad (8)$$

where the unknown vectors $\mathbf{b}_m$ absorb the corresponding powers of h , which are constant within a block. This is a linear system of L equations in the L unknowns $\mathbf{x}_{t_{i-k}}^*, \mathbf{b}_1, \dots, \mathbf{b}_{L-1}$, with coefficient matrix $M \in \mathbb{R}^{L \times L}$ given by

$$M_{n,1} = 1, \quad M_{n,m+1} = S_{p+m-1}(\Lambda_n), \quad m = 1, \dots, L-1. \quad (9)$$

The extrapolated solution is the first component of the solution of that system, so the weights applied to the individual estimates are the first row of M^{-1} . Equivalently, the weight vector $\mathbf{w} \in \mathbb{R}^L$ is obtained from

$$M^\top \mathbf{w} = \mathbf{e}_1, \quad \tilde{\mathbf{x}}_{t_{i-k}} = \sum_{n=1}^L w_n \hat{\mathbf{x}}_{t_{i-k}}^{(k_n)}, \quad (10)$$

where $\mathbf{e}_1$ is the first standard basis vector. Consistency requires $\sum_n w_n = 1$, which follows from the first column of M and is verified numerically in our implementation. For $L = 2$ the matrix reduces to $M = \begin{bmatrix} 1 & 1 \\ 1 & S_p \end{bmatrix}$ and Equation (10) reproduces Equation (7) exactly.

On a uniform grid one has $S_r = n^{1-r}$ for a discretization into n sub-steps, so M becomes a Vandermonde matrix in the nodes $1/n$ and the scheme coincides with the Romberg tableau [12]. On the schedules of Equation (2) this is no longer the case, and the moments must be formed explicitly.

2.4 Nested levels and reuse of predictions

The levels are chosen to be nested, with $k_n = 2^{n-1}$, so that the breakpoints of a coarser level form a subset of those of the next finer level. Nestedness is what keeps the coarse estimates inexpensive: a

coarse sub-step is a union of consecutive fine sub-steps, so a coarse level never visits a time point that the finest trajectory has not already visited, and the stored denoised predictions can be reused in place of fresh evaluations.

At two levels this reuse is exact. The coarse estimate is a single step spanning the block, and its only required prediction is the one at t_i , where the coarse and fine trajectories coincide by construction. This is precisely the property that makes RX-DPM free of additional NFEs.

At three or more levels the situation differs. An intermediate level requires a denoised prediction at a breakpoint that the finest trajectory also visits, but at a different state, because the two trajectories have already separated after the first sub-step. Reusing the stored value therefore introduces an approximation. We retain the state of the coarse level in the drift and approximate only the denoiser output,

$$\mathbf{d} = \frac{\mathbf{x}_{\text{coarse}} - D_{\theta}(\mathbf{x}_{\text{fine}}; t)}{t}, \quad (11)$$

which keeps the dominant part of the drift exact. This is the same trade that the base paper [2] adopts for RX-Runge-Kutta, where an intermediate prediction is approximated by one at a neighbouring time step. As a control we also implement the variant in which every level integrates its own trajectory with fresh evaluations, at the cost of $\sum_{n < L} (k_n - 1)$ additional NFEs per block.

2.5 Conditioning of the weight system

The matrix M of Equation (9) is a generalized Vandermonde matrix, and such matrices are known to be ill conditioned [3]. Two quantities are reported throughout. The first is the condition number $\kappa_{\infty}(M) = \|M\|_{\infty} \|M^{-1}\|_{\infty}$, which bounds the amplification of a relative perturbation of the estimates in the solution. The second is

$$A = \sum_{n=1}^L |w_n|, \quad (12)$$

which we refer to as the cancellation factor. Since the weights sum to unity, A measures directly how far the combination departs from a convex one: a perturbation of size ε in the level estimates may appear as $A\varepsilon$ in the extrapolated value. Weights that alternate in sign with growing magnitude are the signature of a subtraction of nearly equal quantities, and the loss of significant digits it entails.

The system is solved by Gaussian elimination with partial pivoting [3], from which the pivot growth factor $\max |U| / \max |M|$ is also recorded. Reporting the growth factor separates an intrinsic property of the problem from an artefact of the elimination: a growth factor near unity indicates that the pivoting strategy is adequate and that the reported conditioning is a property of M itself.

3 Implementation & Experiments

3.1 An analytically solvable diffusion ODE

Measuring an order of accuracy requires the exact solution, which a learned denoiser never provides. We therefore construct a problem for which Equation (1) is solvable in closed form. If the data distribution is a Gaussian mixture [6], $p_{\text{data}} = \sum_c \pi_c \mathcal{N}(\mu_c, s_c^2 I)$ with weights π_c , means μ_c and standard deviations s_c , then perturbation by noise of scale t leaves the mixture Gaussian, and the exact

denoiser is available:

$$D(\mathbf{x}; t) = \mathbf{x} + t^2 \sum_c \gamma_c(\mathbf{x}) \frac{\mu_c - \mathbf{x}}{s_c^2 + t^2}, \quad (13)$$

where $\gamma_c(\mathbf{x})$ is the posterior responsibility of component c , that is, the normalized weight of that component under the perturbed mixture at $\mathbf{x}$. The ODE being solved is thus the probability flow ODE itself, with an analytic denoiser substituted for the network, and the same sampler code is exercised.

Two instances are used. A single Gaussian gives a drift that is linear in $\mathbf{x}$, for which the flow map admits the closed form

$$\mathbf{x}(\tau_1) - \mu = (\mathbf{x}(\tau_0) - \mu) \sqrt{\frac{s^2 + \tau_1^2}{s^2 + \tau_0^2}}, \quad (14)$$

for any two noise levels τ_0 and τ_1 . This instance therefore provides the most favourable setting for an error model based on step widths alone. A separated bimodal mixture provides a nonlinear drift, for which the ground truth is obtained by a fixed-step fourth-order Runge-Kutta reference solve [15] on a logarithmically spaced grid. The reference is validated against the closed form on the single Gaussian, where it agrees to a maximum relative error of 2.3×10^{-14} , so it is accurate far beyond the errors being resolved.

3.2 Setup for image generation

Image experiments use the EDM backbone [6] with the pretrained conditional CIFAR-10 model [7] at resolution 32×32 , with $t_{\min} = 0.002$, $t_{\max} = 80$ and $\rho = 7$ unless stated otherwise. No training is performed; only the sampler is changed. FID [4] is computed against the standard dataset reference statistics using the Inception network [16] provided with the EDM implementation, so that the numbers are comparable with those reported in the literature.

We generate 10,000 images per FID rather than the 50,000 of the base paper, which is a deliberate concession to the compute budget. FID is a biased estimator and the bias grows as the sample count falls, so our absolute values read higher than published ones. They remain comparable among themselves, since every configuration uses the same sample count and, through per-seed initialization of the latents, the same set of initial conditions. This use of common random numbers is a standard variance reduction technique and is what permits differences smaller than the raw seed to seed spread to be resolved. At the validation gate we additionally report the mean and standard deviation over three disjoint seed blocks.

The baselines are the Euler method, Heun’s method as used by EDM [6, 15], conventional Richardson extrapolation with the fixed coefficients of the uniform-grid formula, and the hybrid RX+EDM in which Heun’s method is applied to the early high noise steps and RX-Euler to the remaining low noise steps. All comparisons are made at matched NFE rather than at matched step count, since Heun’s method requires two evaluations per step while RX-Euler requires one. Heun’s method can only realize odd NFE, so its points appear one above the even values of the grid, following the convention of the base paper.

Table 1: Fitted global order of accuracy on the bimodal problem, obtained by least squares over the asymptotic region. All fits have $r^2 \geq 0.96$.

Method	Fitted order	Predicted
Euler	1.02	1
Heun (EDM)	2.05	2
Conventional Richardson	1.04	2
RX-Euler ($L = 2$)	1.94	2
RX ($L = 3$)	1.72	3
RX ($L = 4$)	1.88	4

3.3 Verification against the reference implementation

Our sampler is an independent reimplementaion, so its agreement with the published one is established before any result is interpreted. The reference sampler and its coefficient routine are transcribed verbatim from the official release and compared against ours on the analytic problem, which is deterministic and requires no GPU. The two agree to a maximum absolute difference below 10^{-12} across eight combinations of step count and block length, with identical NFE in every case, and our general L -level solver reduces to the published closed form at $L = 2$ to machine precision.

3.4 Numerical procedures

All samplers are implemented in PyTorch [10]. The weight system of Equation (10) is solved by our own implementation of LU factorization with partial pivoting [3], which also supplies the condition number and the growth factor. Orders of accuracy are extracted by least squares fitting of $\log E$ against $\log h$ or $\log N$, where E is the measured error, h the block width and N the number of steps; the fit is performed through the normal equations [1]. The schedule exponent ρ is tuned by golden-section search [1], which requires no derivative and makes the fewest objective evaluations of any bracketing method for a given final interval, a relevant consideration when each evaluation is a complete FID measurement. The extrapolation arithmetic is additionally carried out in single and half precision to isolate the contribution of round-off, with the trajectory itself integrated in double precision throughout so that any measured effect is attributable to the weight solve and the combination alone.

4 Results & Discussion

4.1 Verification of the error model

Figure 1 and Table 1 report the global order of accuracy measured on the analytic problem. The orders are fitted over the asymptotic region, by which we mean the larger step counts, where the leading term of the expansion dominates and the fitted slope is meaningful. At the smallest step counts an L -level block spans so much of the noise range that the expansion underlying Equation (8) does not apply, and the extrapolated samplers are then visibly worse than the Euler baseline, as the upper left of each panel shows. The Euler method and Heun’s method attain their textbook orders of 1.02 and 2.05, which confirms that the harness itself is sound. RX-Euler attains 1.94, consistent with the order of $O(N^{-2})$ claimed for it.

Table 2: FID at the validation gate, reported as mean and standard deviation over three disjoint seed blocks. Our values use 10,000 images and the published values use 50,000, so the two columns are not directly comparable in magnitude.

Method	NFE	Ours	Base paper
Euler	10	17.48 ± 0.30	15.88
Heun (EDM)	11	16.45 ± 0.15	14.46
RX-Euler	10	6.15 ± 0.04	4.35
RX+EDM	10	10.72 ± 0.07	4.26

Table 3: FID on CIFAR-10 for conventional Richardson extrapolation with fixed coefficients against the grid-aware coefficients of RX-DPM, and the effect of the block length k .

NFE	6	8	10	12	16	20
Euler	38.43	23.78	17.14	13.58	9.89	8.06
Conventional Richardson	30.21	18.30	13.41	10.65	7.91	6.60
RX-Euler ($k = 2$)	33.27	8.21	6.12	5.47	4.86	4.57
RX-Euler ($k = 3$)	22.99	12.69	9.04	5.95	5.21	4.94
RX-Euler ($k = 4$)	33.54	9.70	10.29	7.90	5.43	4.96

Conventional Richardson extrapolation, by contrast, remains at order 1.04 and does not reach the order it would have on a uniform grid, which is a direct consequence of applying fixed coefficients to a non-uniform schedule.

The fitted orders for $L = 3$ and $L = 4$ are 1.72 and 1.88, well short of the predicted 3 and 4. The extension does not deliver the additional orders that Equation (8) anticipates. The remainder of this section establishes the cause.

4.2 Reproduction on CIFAR-10

Figure 2 and Table 2 present the reproduction. RX-Euler reduces FID from 17.48 to 6.15 at NFE = 10, a factor of 2.8, and it also improves on Heun’s method even though Heun is granted one additional evaluation. The ordering of the four methods agrees with the base paper. The absolute values lie above the published ones for the sample size reason stated in Section 3.2, and the spread over seed blocks in Figure 3 is small enough that every difference discussed here is well outside the sampling variability. Consistent with the base paper, RX-Euler is strongest in the fast sampling regime and Heun’s method closes the gap as the budget increases.

The one point of disagreement concerns RX+EDM, which we find to be worse than plain RX-Euler, at 10.72 against 6.15, whereas the base paper reports it to be slightly better. The division of the trajectory between Heun steps and RX-Euler steps is not fully specified in the paper; we allocate the first third of the budget to Heun. We therefore attribute the discrepancy to this choice rather than treating it as a refutation.

4.3 Grid-aware against fixed coefficients

Figure 4 and Table 3 reproduce the ablation of the base paper. From NFE = 8 upward the grid-aware coefficients improve on the fixed coefficients of conventional Richardson extrapolation at every budget,

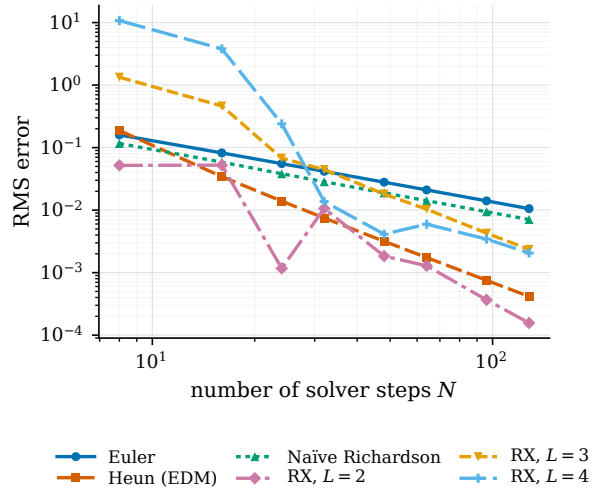
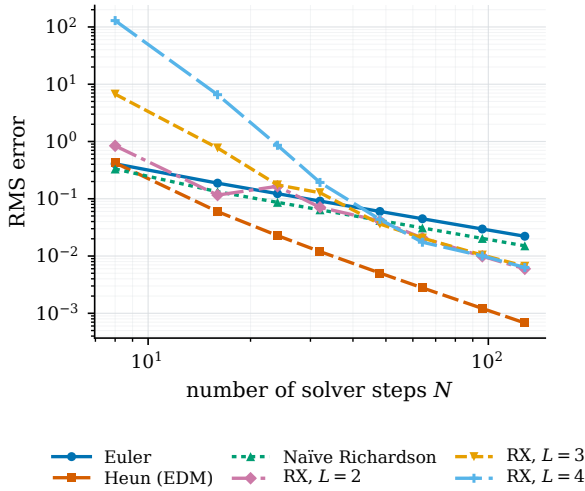


Figure 1: Global RMS error against the number of solver steps on the analytic problem, for the bimodal mixture (left) and the single Gaussian (right). The extrapolated samplers improve on the Euler method, but the curves for $L = 3$ and $L = 4$ do not separate from that for $L = 2$ in the asymptotic region.

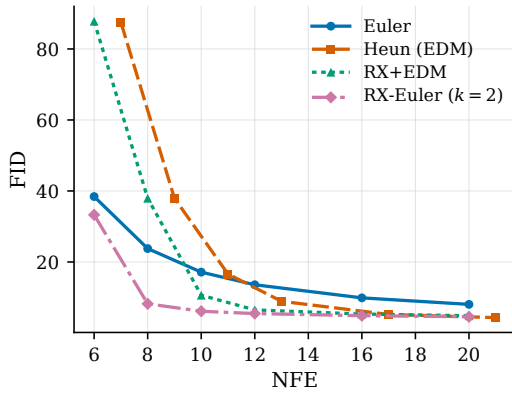


Figure 2: FID against NFE on conditional CIFAR-10 with the EDM backbone. RX-Euler dominates at low NFE, and Heun’s method approaches it only as the budget grows.

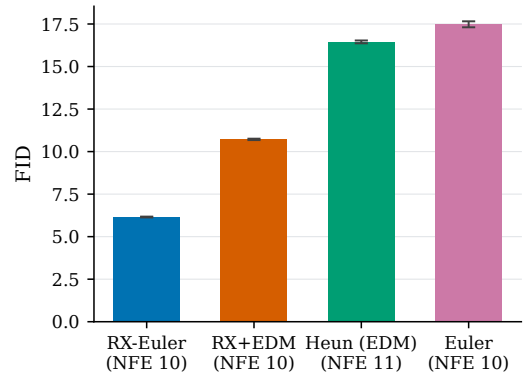


Figure 3: FID at the validation gate with the spread over three disjoint seed blocks shown as the standard error of the mean. The separation between the methods is far larger than the sampling variability.

Table 4: FID on CIFAR-10 against the number of extrapolation levels, with the block length set to $k = 2^{L-1}$.

NFE	$L = 2$	$L = 3$	$L = 4$
8	8.21	242.8	153.5
10	6.12	66.3	113.2
16	4.86	65.2	80.6

by a factor of 2.2 at $\text{NFE} = 10$. The single exception is the smallest budget, $\text{NFE} = 6$, where $k = 2$ falls behind the fixed coefficients at 33.27 against 30.21; the blocks are then so wide that the leading term of the expansion no longer dominates, and the grid-aware coefficients lose the justification on which they rest. The best block length at that budget, $k = 3$, still reaches 22.99 and remains ahead. This confirms empirically that coefficients adapted to arbitrary time step scheduling are more effective than deterministic ones derived under uniform discretization. Among the block lengths, $k = 2$ is the best choice over most of the range, in agreement with the setting adopted by the base paper, and the curves for $k \geq 3$ remain closer to $k = 2$ than to the Euler baseline, so even sparse extrapolation retains much of the benefit.

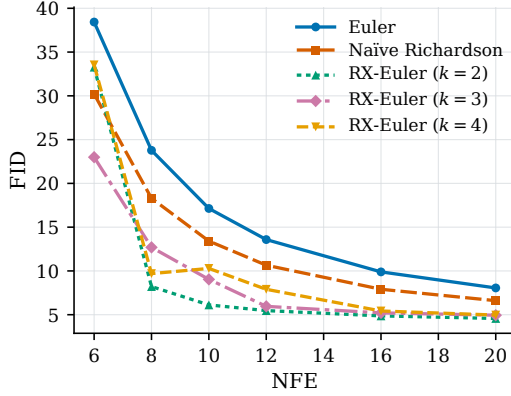


Figure 4: Effect of the extrapolation coefficients and of the block length k on the Euler method. The grid-aware coefficients improve on the fixed ones at every budget, and $k = 2$ is the strongest setting.

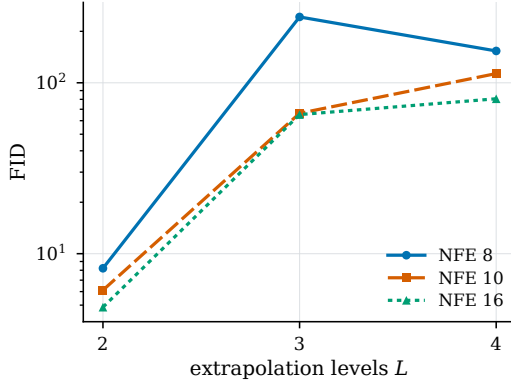


Figure 5: FID against the number of extrapolation levels, at three budgets. The vertical axis is logarithmic. Two levels is the best setting at every budget.

4.4 The multi-level extension

Figure 5 and Table 4 show the behaviour of the extension on CIFAR-10. Adding a third level raises FID from 6.12 to 66.3 at NFE = 10, an order of magnitude in the wrong direction, and a fourth level is worse still. The failure observed on the analytic problem in Section 4.1 is therefore not an artefact of that problem; it transfers unchanged to a real network and a perceptual metric.

4.5 The origin of the failure

To isolate the cause we measure the local truncation error of a single block from an exact initial condition, so that the measurement is uncontaminated by accumulation across blocks, and shrink the block to extract an order. Figure 6 and Table 5 give the result, which is consistent on both the linear and the nonlinear problem.

At $L = 2$ the reuse and exact variants agree exactly, at 3.22 and 3.22. This is the expected behaviour, since nothing is approximated at two levels. For $L \geq 3$ the reuse variants saturate near 3.2 regardless of the number of levels, while exact recomputation follows the

Table 5: Fitted local order of a single extrapolation block on the bimodal problem, with $p = 2$ and $r^2 \approx 0.999$ throughout. The final column gives the additional NFEs per block required by exact recomputation.

L	Reuse	Exact	Predicted $p + L - 1$	Extra NFE
2	3.22	3.22	3	0
3	3.21	4.28	4	1
4	3.22	5.36	5	4
5	3.22	5.90	6	11

Table 6: Conditioning of the weight system against the number of levels, with $k = 2^{L-1}$ on the EDM schedule at $\rho = 7$.

L	$\kappa_\infty(M)$	$A = \sum_n w_n $	Growth factor
2	8.0	3.00	1.0
3	6.0×10^1	5.01	1.0
4	5.8×10^2	6.50	1.0
5	7.4×10^3	7.51	1.0
6	9.8×10^4	8.28	1.0
7	6.1×10^5	8.99	1.0

predicted order $p + L - 1$ closely, reaching 4.28, 5.36 and 5.90 against the predicted 4, 5 and 6. The agreement of the exact variant with the prediction shows that the error model of Equation (8) and the linear algebra built on it are sound; it is the reuse of stored predictions, and not the extrapolation itself, that limits the attainable order.

The consequence is that the property which makes RX-DPM free of additional NFEs is specific to two levels. Buying further orders requires buying further evaluations, and Figure 7 shows that on CIFAR-10 this exchange is not profitable. Three levels with exact recomputation reach an FID of 6.35 at NFE = 20, whereas two levels with reuse reach 4.57 at the same budget.

4.6 Conditioning of the weight system

Table 6 and Figure 8 quantify the conditioning. The condition number grows by roughly one order of magnitude per level, from 8.0 at two levels to 6.1×10^5 at seven, and the weights alternate in sign with growing magnitude, which is the expected signature of cancellation. The growth factor remains at 1.0 throughout, so the ill conditioning is a property of the moment system itself and not of the elimination used to solve it.

The lower right panel of Figure 8 separates two effects that might otherwise be conflated. Holding the number of levels fixed and varying the block length leaves the condition number nearly unchanged, and in some cases slightly improves it, so the growth reported in Table 6 is attributable to the number of levels and not to the wide spread of step widths that a long block on a geometric schedule produces.

The lower left panel reports the accuracy of the weights themselves. In single precision the relative error remains below 10^{-3} up to seven levels, while in half precision it reaches 0.23 at five levels and 0.92 at seven, at which point the computed weights carry no useful information. In both cases the measured error stays below the ε_K bound, where ε is the unit round-off of the format.

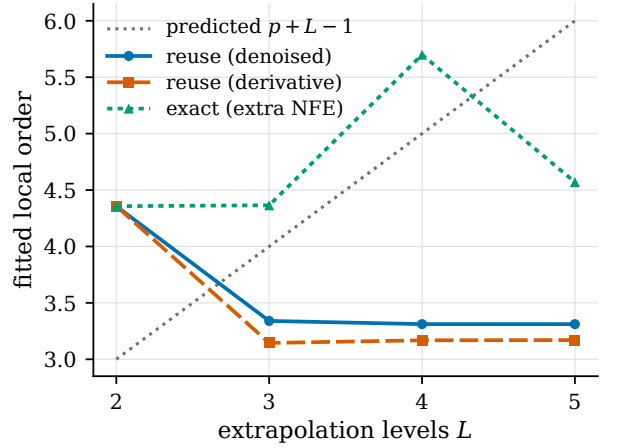
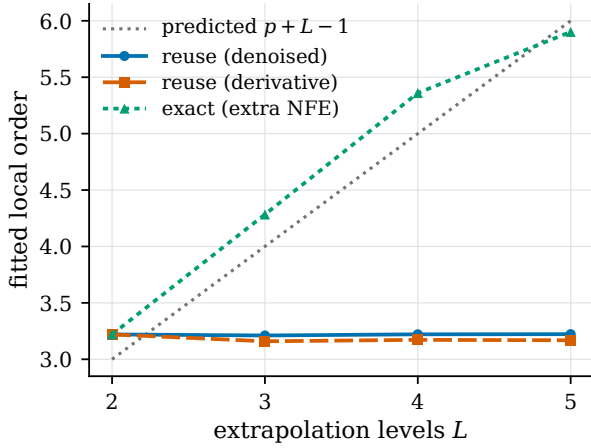


Figure 6: Fitted local order of a single extrapolation block against the number of levels, for the bimodal mixture (left) and the single Gaussian (right). The reuse variants coincide with exact recomputation at $L = 2$ and then saturate, while exact recomputation follows the predicted order.

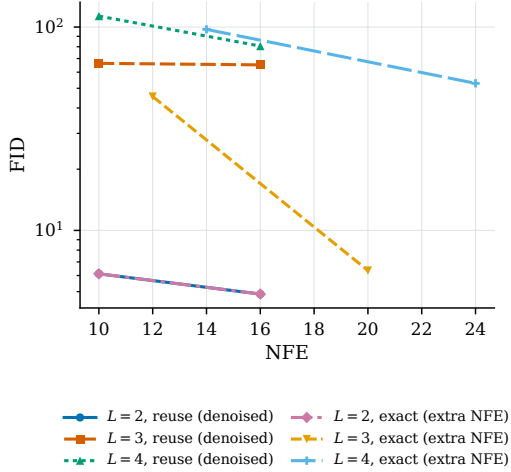


Figure 7: FID against NFE for the reuse and exact variants. The two coincide at $L = 2$, and neither is competitive at higher level counts.

The conditioning reported here is a property of the weight system and not, by itself, a statement about sample quality. Whether it matters depends on how it compares with the truncation error that remains, which the next section examines.

4.7 Truncation error against round-off

The upper row of Figure 9 exhibits the classical balance between truncation error and round-off error. In double precision at two levels the error decreases with refinement until $N = 256$, where it reaches 1.53×10^{-6} , and then turns upward, reaching 3.81×10^{-6} at $N = 2048$. Single precision tracks double precision almost exactly. Half precision behaves differently: it saturates near 10^{-3} and no amount of refinement improves it, so reduced precision manifests as an error floor rather than as a displaced minimum.

The practical consequence is that round-off is not the operative limit for this application. The minimum of the double precision curve lies at $N = 256$, whereas samplers of interest operate at N between 10 and 20, that is, deep inside the region where truncation error dominates. The lower right panel of Figure 9 confirms this on CIFAR-10: at the only usable setting, $L = 2$, moving the extrapolation arithmetic to single precision changes FID by less than 0.1% and to half precision by at most a few tenths of a percent. The differences that do appear arise only at $L \geq 3$, where the sampler is already far from competitive.

The lower left panel of Figure 9 completes the picture. Sweeping the number of levels produces no minimum at all; the error increases monotonically in every precision tested. This follows directly from the reuse ceiling of Section 4.5. Since the additional levels purchase no additional order, their only effect is to introduce the conditioning cost documented in Table 6.

4.8 The schedule exponent

The exponent ρ of Equation (2) was selected by EDM for the Euler and Heun samplers. Since the extrapolation weights are computed from the step widths, ρ also determines the conditioning of the weight system, and there is no reason for a single value to be simultaneously optimal for both. Figure 10 examines this. On the analytic problem a dense scan places the optimum at $\rho = 6.25$ for the Euler method, close to the value adopted by EDM and a useful check on the harness, and at $\rho = 8.0$ for RX-Euler. On CIFAR-10 the golden-section search settles at $\rho = 6.16$ with an FID of 5.95, against 6.12 at $\rho = 7$, an improvement of 2.7%.

The gain is real but modest, and the objective is not cleanly unimodal, as the scan in the left panel shows. The appropriate conclusion is that the favourable region lies near $\rho = 6$ rather than that a particular minimizer has been certified. This is also why the dense scan is reported alongside the search: golden-section search presumes unimodality, and it is better to exhibit the objective than to assume the property.

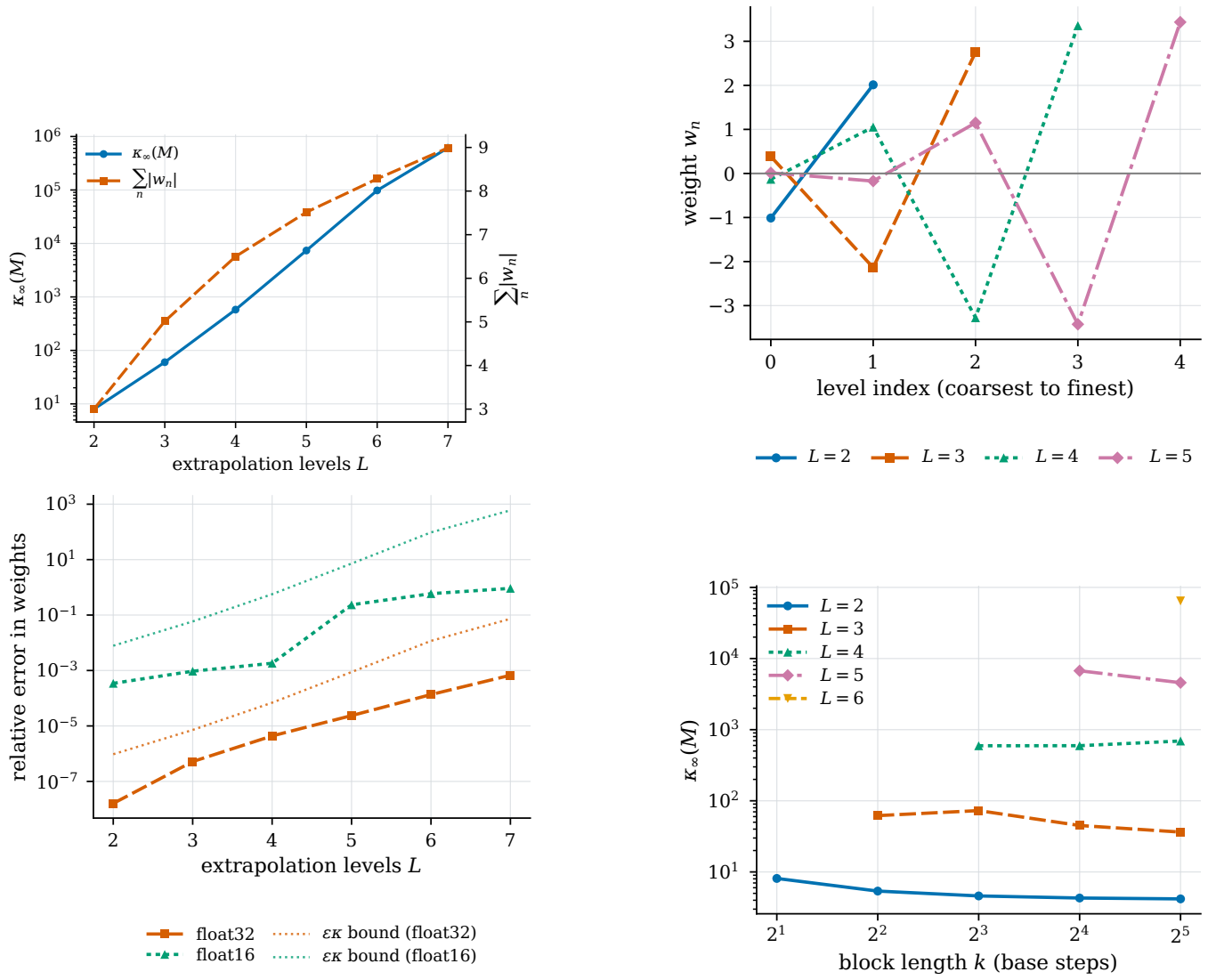


Figure 8: Conditioning of the weight system. Upper left: condition number and cancellation factor against the number of levels. Upper right: the weights themselves, which alternate in sign and grow in magnitude as levels are added. Lower left: relative error of the computed weights, with the corresponding $\epsilon\kappa$ bounds shown dotted. Lower right: condition number against the block length k , which shows that the conditioning is governed by the number of levels rather than by the spread of the step widths.

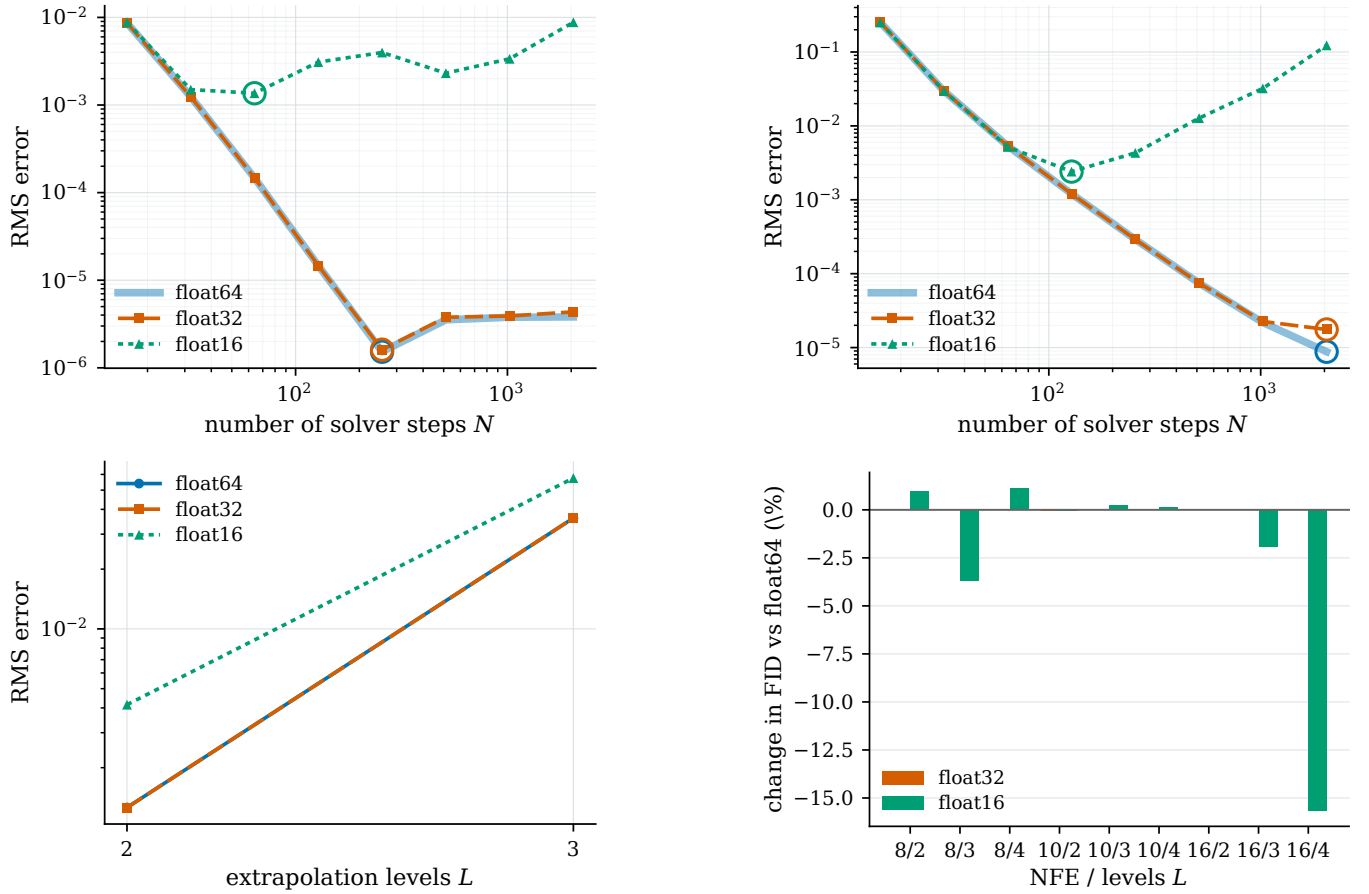


Figure 9: Truncation error against round-off error. Upper row: RMS error against the number of solver steps at two levels (left) and three levels (right), for three working precisions of the extrapolation arithmetic, with open circles marking the minimum of each curve. Lower left: RMS error against the number of levels, which increases monotonically. Lower right: change in FID relative to double precision on CIFAR-10, by budget and level count.

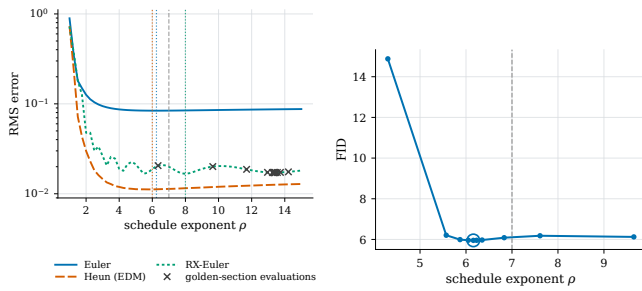


Figure 10: Left: RMS error against the schedule exponent ρ on the analytic problem, with the evaluations of the golden-section search overlaid and the value $\rho = 7$ marked. Right: FID against ρ on CIFAR-10 at the evaluated points.

5 Conclusion & Future Work

We reproduced RX-DPM on CIFAR-10 with the EDM backbone and extended it from two extrapolation levels to L nested levels, in the manner of a Romberg tableau, by solving a generalized Vandermonde moment system for the weights. The reproduction confirms the claims of the base paper. RX-Euler lowers FID from 17.48 to 6.15 at $NFE = 10$ and improves on Heun’s method at matched cost, and the grid-aware coefficients improve on the fixed coefficients of conventional Richardson extrapolation at every budget but the smallest, which establishes that adapting the coefficients to the actual step widths is what makes the method effective on the non-uniform schedules that DPMs use.

The extension does not succeed, and the cause is specific and measurable. The reuse of stored denoised predictions, which is exactly the mechanism that makes RX-DPM free of additional NFEs, is exact only at two levels, where the coarse and fine trajectories still coincide at the single point the coarse estimate requires. For $L > 2$ it limits the attainable local order to approximately 3.2 regardless of the number of levels, whereas exact recomputation follows the predicted order $p + L - 1$ closely. The agreement of the exact variant with the prediction shows that the error model and the linear algebra built on it are sound, and that it is the reuse alone that is responsible. Since exact recomputation costs more evaluations than it returns, the conclusion is that the free-of-charge property of RX-DPM is a property of two levels and does not survive generalization.

Two secondary findings support this. The conditioning of the weight system degrades by roughly one order of magnitude per level, from $\kappa_\infty = 8.0$ at two levels to 6.1×10^5 at seven, while the pivot growth factor remains at unity, so the degradation is intrinsic to the moment system. Round-off, by contrast, is not the operative limit for this application: the minimum of the double precision error curve lies near $N = 256$, whereas samplers of interest operate between $N = 10$ and $N = 20$, and moving the extrapolation arithmetic to half precision changes FID by at most a few tenths of a percent at two levels.

Limitations. FID is computed from 10,000 images rather than 50,000, so our absolute values are biased upward and are comparable among themselves rather than against published numbers. The image experiments use a single dataset and backbone, and the extension is evaluated only with the Euler method as the baseline solver, that is, for $p = 2$. The nested construction restricts the level counts to powers of two. Finally, our RX+EDM results depend on a division between Heun steps and RX-Euler steps that the base paper does not fully specify, and we attribute our disagreement on that configuration to this choice rather than to the method itself.

Future work. The most direct continuation is to relax the assumption of linear error accumulation in Equation (8). The exact variant reaches the predicted order under that assumption, so an error model that also captures the terms arising from propagation through the Jacobian of the drift, which the moments of the step widths alone cannot express, may permit coefficients that tolerate the reuse approximation. A second direction is to select the number of levels adaptively along the trajectory rather than globally, since the block geometry and the accuracy of the predictions both

vary with the noise level. Applying the multi-level construction to higher-order baseline solvers such as DPM-Solver and PNDM, for which $p > 2$ and the reuse pattern differs, would establish whether the ceiling reported here is particular to the Euler method. Finally, the observation that the favourable region for the schedule exponent lies near $\rho = 6$ for RX-Euler rather than at the value of 7 adopted for Euler and Heun suggests that the schedule and the extrapolation coefficients are better tuned jointly than separately.

References

- [1] Steven C. Chapra and Raymond P. Canale. 2021. *Numerical Methods for Engineers* (8 ed.). McGraw-Hill Education, New York, NY.
- [2] Jinyoung Choi, Junoh Kang, and Bohyung Han. 2025. Enhanced Diffusion Sampling via Extrapolation with Multiple ODE Solutions. In *International Conference on Learning Representations (ICLR)*. arXiv:2504.01855 [cs.LG]
- [3] Gene H. Golub and Charles F. Van Loan. 2013. *Matrix Computations* (4 ed.). Johns Hopkins University Press, Baltimore, MD.
- [4] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. In *Advances in Neural Information Processing Systems (NIPS)*. 6626–6637. arXiv:1706.08500 [cs.LG]
- [5] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. 6840–6851. arXiv:2006.11239 [cs.LG]
- [6] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. 2022. Elucidating the Design Space of Diffusion-Based Generative Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2206.00364 [cs.CV]
- [7] Alex Krizhevsky. 2009. *Learning Multiple Layers of Features from Tiny Images*. Technical Report. University of Toronto, Toronto, Ontario.
- [8] Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. 2022. Pseudo Numerical Methods for Diffusion Models on Manifolds. In *International Conference on Learning Representations (ICLR)*. arXiv:2202.09778 [cs.CV]
- [9] Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. 2022. DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2206.00927 [cs.LG]
- [10] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. 2019. PyTorch: An Imperative Style, High-Performance Deep Learning Library. In *Advances in Neural Information Processing Systems (NeurIPS)*. 8024–8035.
- [11] Lewis Fry Richardson. 1911. The Approximate Arithmetical Solution by Finite Differences of Physical Problems Involving Differential Equations, with an Application to the Stresses in a Masonry Dam. *Philosophical Transactions of the Royal Society of London. Series A* 210, 459–470 (1911), 307–357. doi:10.1098/rsta.1911.0009
- [12] Werner Romberg. 1955. Vereinfachte numerische Integration. *Det Kongelige Norske Videnskabers Selskab Forhandling* 28, 7 (1955), 30–36.
- [13] Jiaming Song, Chenlin Meng, and Stefano Ermon. 2021. Denoising Diffusion Implicit Models. In *International Conference on Learning Representations (ICLR)*. arXiv:2010.02502 [cs.LG]
- [14] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021. Score-Based Generative Modeling through Stochastic Differential Equations. In *International Conference on Learning Representations (ICLR)*. arXiv:2011.13456 [cs.LG]
- [15] Endre Süli and David F. Mayers. 2003. *An Introduction to Numerical Analysis*. Cambridge University Press, Cambridge.
- [16] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. 2016. Rethinking the Inception Architecture for Computer Vision. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2818–2826.